

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 20

Code of Conduct

I _Narmatha J (192521341)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _Narmatha J_

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

CO2 – ASSESSMENT TOOL 2

ANALYTICAL PROBLEM SOLVING

Q1. Design a Presence Management Solution for a Collaborative Online Learning Platform.

Introduction:

Presence management is a technology that monitors and displays the real-time availability and activity status of users in a communication system. In a collaborative online learning platform, it enables students and teachers to know who is online, attending a live class, busy, or offline. This information improves communication, collaboration, and the overall learning experience by ensuring that users can interact with the right person at the right time.

Problem Statement:

In online learning platforms, communication delays often occur because students and teachers are unaware of each other's availability. Students may send messages when teachers are unavailable, and group discussions may be delayed because members are offline. Therefore, an efficient presence management system is required to provide real-time user status and improve collaboration.

Proposed Presence Management Solution:

1. User Authentication:

Every student and teacher must log in using a valid username and password. The authentication module verifies the user's identity before allowing access to the platform.

2. Presence Detection Module:

The system continuously monitors user activity. Based on user behaviour, it automatically updates the presence status.

Example statuses include:

- Online
- Offline
- Away
- Busy
- In Class
- Available for Chat

3. Presence Server:

The Presence Server acts as the central component of the system. It receives status updates from all users, stores the current status in the database, and distributes updates to authorised users in real time.

4. Database Management:

The database stores user profiles, login information, presence status, last active time, and notification history. This information is updated continuously.

5. Notification Service:

Whenever a user's status changes, the notification service informs other users instantly. For example, when a teacher becomes online, students receive a notification that the teacher is available.

6. User Dashboard:

The dashboard displays:

- Online students
- Online teachers
- Students attending live classes
- Last seen information
- Current presence status

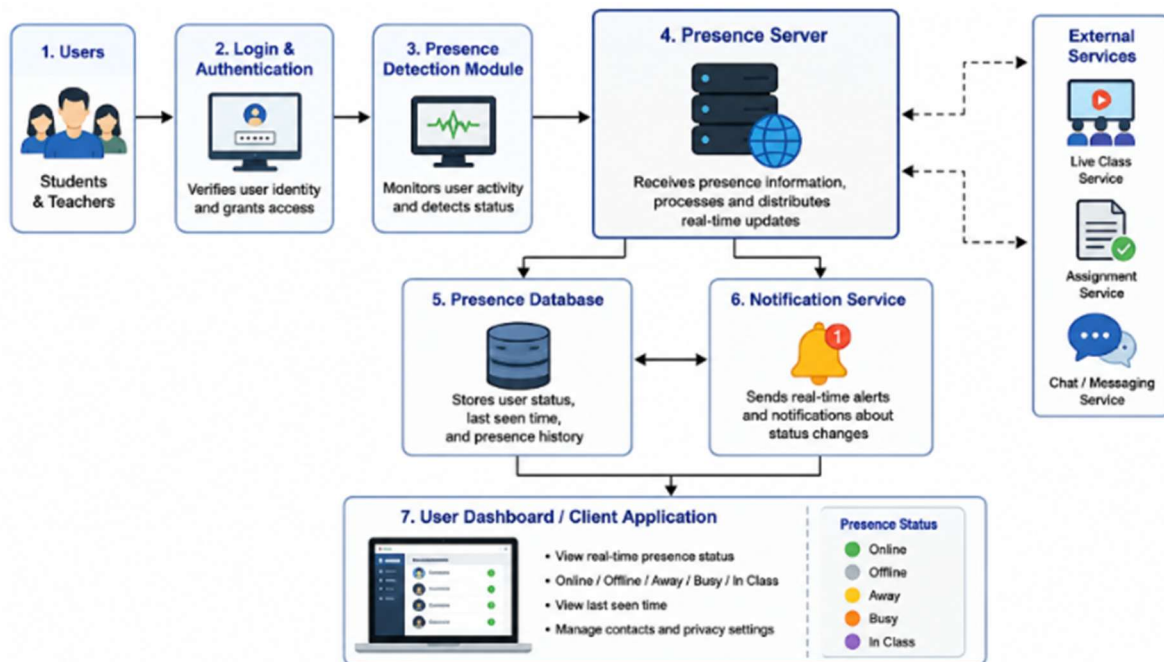
7. Privacy Management:

Users can control who can view their presence information. They may choose to share their status with everyone, only teachers, only classmates, or selected users.

Working Process:

- User logs into the platform.
- Authentication verifies the user.
- The Presence Detection Module monitors user activity.
- The Presence Server receives the updated status.
- The Database stores the latest information.
- Notifications are sent to authorised users.
- The dashboard displays the updated presence information.

System Architecture:



Advantages:

- Provides real-time user availability.
- Improves communication between teachers and students.
- Supports collaborative learning.
- Reduces communication delays.
- Enables faster decision-making.
- Enhances the overall learning experience.

Conclusion:

The proposed presence management solution enables real-time monitoring of user availability, making communication faster and more efficient. By integrating authentication, presence detection, notifications, and privacy controls, the system creates a collaborative and interactive learning environment.

Q2. Evaluate How Presence Information Improves Communication Efficiency

Introduction:

Presence information refers to the real-time status of users in a communication system. It indicates whether a user is online, offline, busy, away, or available. In collaborative online learning, presence information improves communication by helping users identify the availability of teachers and classmates before initiating communication.

Role of Presence Information:

1. Faster Communication:

Users can immediately identify who is available and contact them without waiting unnecessarily.

2. Reduces Communication Delays:

Instead of sending messages to offline users, students can communicate with users who are currently online, resulting in quicker responses.

3. Improves Collaboration:

Group members can identify active participants and complete assignments more efficiently through real-time communication.

4. Supports Live Learning:

Teachers can conduct live classes, answer questions instantly, and provide immediate feedback because they know which students are online.

5. Better Resource Utilisation:

Teachers can schedule discussions and meetings when most students are available, improving classroom participation.

6. Increases Productivity:

Users spend less time searching for available contacts and more time focusing on learning activities.

7. Enhances User Experience:

Real-time status indicators make the platform more interactive and user-friendly.

Analytical Evaluation:

Without Presence Information	With Presence Information
Users do not know who is available.	Users can identify available contacts instantly.
Communication is delayed.	Faster communication and response.
More failed communication attempts.	Fewer unnecessary messages and calls.
Collaboration is difficult.	Group work becomes more effective.
Lower productivity.	Higher productivity and engagement.

Challenges:

- Privacy concerns if status is always visible.
- Incorrect status updates due to network issues.
- Increased server load because of continuous status monitoring.
- Requires secure authentication and data protection.

Conclusion:

Presence information greatly improves communication efficiency by providing real-time availability information, reducing delays, enhancing collaboration, and supporting interactive online learning. Although there are privacy and technical challenges, the benefits make presence management an essential component of modern collaborative learning systems.